INCH-POUND

MIL-M-38510/346B
AMENDMENT 2
6 November 1992
SUPERSEDING
AMENDMENT 1
27 October 1987

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
TRANSPARENT AND OCTAL D TYPE LATCHES, CASCADABLE,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/346B, dated 12 February 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2.3, add the following new case outline:

"X C-2A (20 terminal, .350" x .350"), square chip carrier package".

* 1.3, thermal resistance: Delete as follows (alter change made by amendment 1):

"Thermal resistance, junction-to-case (Θ_{JC}):
Cases R, S - (See MIL-M-38510, appendix C)
Case 2 - 60°C/W 2/"

Substitute:

"Thermal resistance, junction-to-case (Θ_{JC}):
Cases R, S, X, and 2 - - - - - - - - - - - - - - - (See MIL-M-38510, appendix C)"

* Footnote 2/: Delete in its entirety.

PAGE 3

TABLE I, low-level input current, minimum limits column: Delete "-0.03" and substitute "0.0".

PAGE 4

TABLE I, LE to Q, t_{PLH2}, device type 01, maximum limits column: Delete "15.5" and substitute "15.0".

* TABLE I, $\overline{OE}$ to $\overline{Q}$, t_{PLZ1}, device type 02, minimum limits column: Delete "2.0" and substitute "1.5".

* TABLE I, $\overline{OE}$ to $\overline{Q}$, t_{PHZ1}, device type 02, minimum limits column: Delete "2.0" and substitute "1.5".

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FIGURE 1: Delete "Case 2" and substitute "Cases 2 and X" (two places).

PAGE 10

FIGURE 1: Delete "Case 2" and substitute "Cases 2 and X".

PAGE 11

FIGURE 1: Delete "Case 2" and substitute "Cases 2 and X".

PAGE 12

FIGURE 2, device types 01 and 02: Delete "OE" and substitute "$\overline{OE}$" (two places).

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* FIGURE 3, Device types 01 and 04: Delete "OE" and "Qn" and substitute "$\overline{OE}$" and "Q_n", respectively. Device type 02: Delete "OE" and "Qn" and substitute "OE" and "Q_n", respectively. (alter change made by amendment 1).

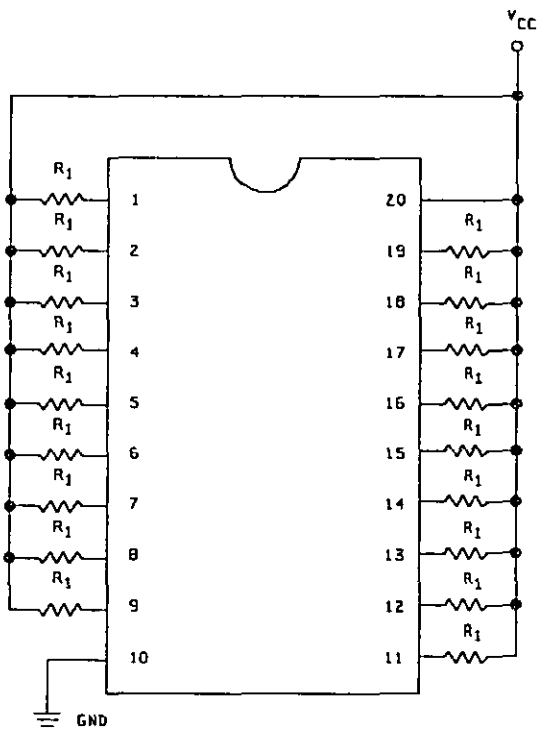
PAGES 17 THROUGH 20

FIGURE 4: Delete "OE" and substitute "$\overline{OE}$".

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* Following figure 4, add new circuit and notes as follows (alter change made by amendment 1):

"



NOTES:
1. High temperature condition F (+175°C) burn-in circuit only.
2. R_1 = 2 kΩ.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 4. Burn-in and life test circuit - Continued."

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TABLE III, column heading: Delete "Cases R, S, 2" and substitute "Cases R, S, X, 2".

PAGE 24

TABLE III, $\overline{OE}$ pin, test numbers 82 through 88: Add "„" (seven places).

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* TABLE III, device type 01, footnote 1/: Delete and substitute: "1/ A ≥ 2.5 V, B ≤ 0.5 V".

TABLE III, device type 01, footnote 4/: Delete and substitute as follows:

"4/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0.0/-.30

"

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TABLE III, device type 02, $\overline{OE}$ pin, test numbers 82 through 88: Add "„" (seven places).

PAGE 30

* TABLE III, device type 02, t_{PHZ1}, test number 133, minimum limits column: Delete "2.0" and substitute "1.5".

TABLE III, measured terminal column, test numbers 133 through 164: Delete "$Q_0...Q_7$" and substitute "$\overline{Q}_0...\overline{Q}_7$" as appropriate (32 places).

* TABLE III, device type 02, t_{PZH1}, test number 149, minimum limits column: Delete "„" and substitute "2.0".

* TABLE III, device type 02, footnote 1/: Delete and substitute: "1/ A ≥ 2.5 V, B ≤ 0.5 V".

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TABLE III, LE pin, test number 9: Delete "1.1 V" and substitute "„".

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TABLE III, GND pin, test number 82: Add "„".

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TABLE III, measured terminal column, test numbers 140 through 155: Delete "$Q_0...Q_7$" and substitute "$\overline{Q}_0...\overline{Q}_7$" as appropriate (16 places).

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TABLE III, LE pin, test number 9: Delete "1.1 V" and substitute "„".

PAGE 36

TABLE III, GND pin, test number 82: Add "„".

PAGE 39

* TABLE III, device type 04, footnote 3/: Delete and substitute: "3/ A ≥ 2.5 V, B ≤ 0.5 V, H ≥ 2.5 V, L ≤ 0.5 V".

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TABLE IV, delete and substitute:

"TABLE IV. Manufacturers' designations.

Device type	Circuits			
	A	B	C	D
	Fairchild Semiconductor	Motorola	Signetics	Texas Instruments
01	X	X	X	X
02	X	X		
03	X			
04	X			

".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, correction, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1297)

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